Dual Drive RF Amp Replacement

(Click Here for Summary Version)

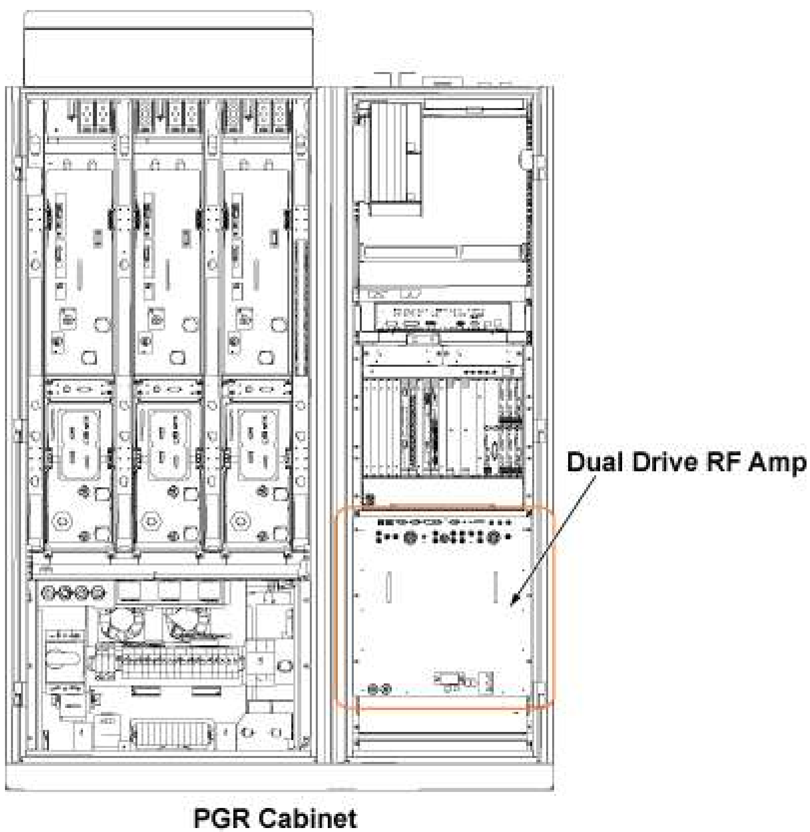
1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	5 mins	90-120	5 mins

2 Overview

This document contains Dual Drive RF Amp replacement procedure. This unit is in the PGR cabinet (shown below). The Dual Drive RF Amp unit is about 335 lbs (152kg). The unit weighs enough that the use of a hoist kit is required. This document details the correct way to use a hoist kit to remove the Dual Drive RF Amp unit and replace with the field replaceable unit (FRU) safely.

Illustration 1: Dual Drive RF Amp



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Hoist Service Kit	1	-	5196226	-
Non-Magnetic Tool Kit (or equivalent)	1	-	5113258	-

3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Towels	N/A	-	-	-
Tie Wraps	3-5	-	-	-

3.3 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Dual Drive RF Amp	1	-	See FRU manual	-

3.4 Safety

DANGER

ELECTROCUTION HAZARD!
HIGH VOLTAGE PRESENT.
USE PROPER LOCK OUT / TAG OUT PROCEDURES BEFORE SERVICING AND/OR MAINTAINING.

WARNING

POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE!
THE MODULE COULD FALL WHEN USING THE LIFT TOOL.
BEFORE PUTTING WEIGHT ON THE HOIST CHAIN, BE SURE THAT THE LINKS ARE ALIGNED PROPERLY. THIS WILL HELP KEEP THE CHAIN FROM BINDING UP, OR BECOMING KINKED, WHICH IS NEARLY IMPOSSIBLE TO CORRECT WHEN THE CHAIN IS UNDER TENSION.

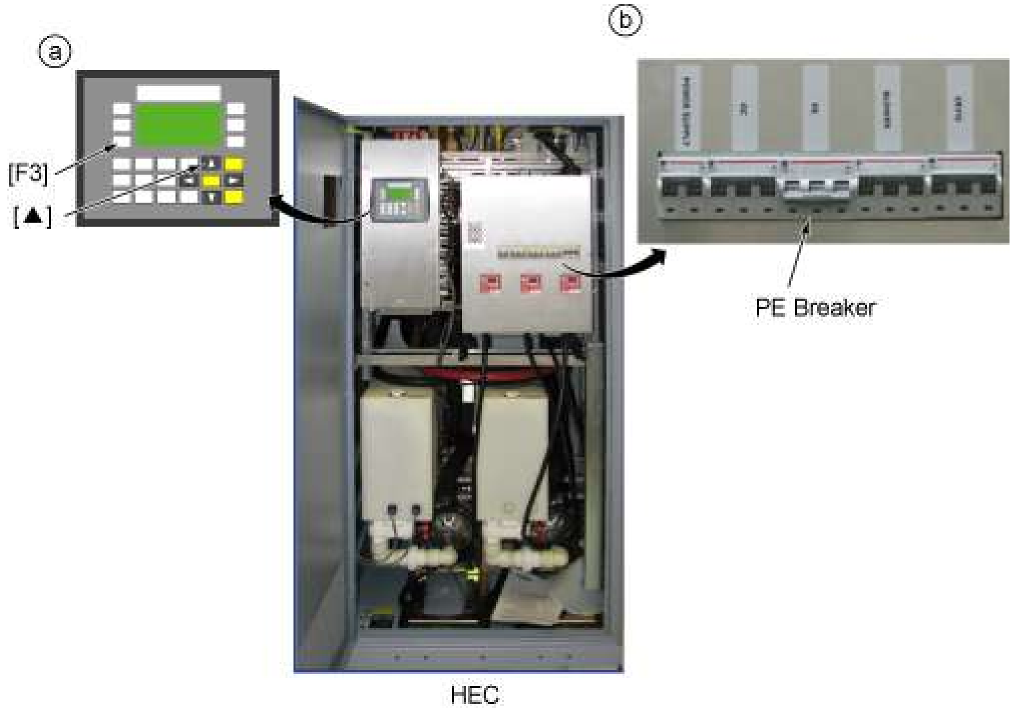
3.5 Required Conditions

Condition	Reference	Effectivity
The PGR PDU/Gradient Subsystem must be locked out and tagged out before starting this procedure.	-	-
Before disconnecting any coolant lines on the front of the Dual Drive RF Amp unit, shut down the Power electronics PuMP (PPMP) at the Heat Exchange Cabinet (HEC) .	-	-

4 Procedure

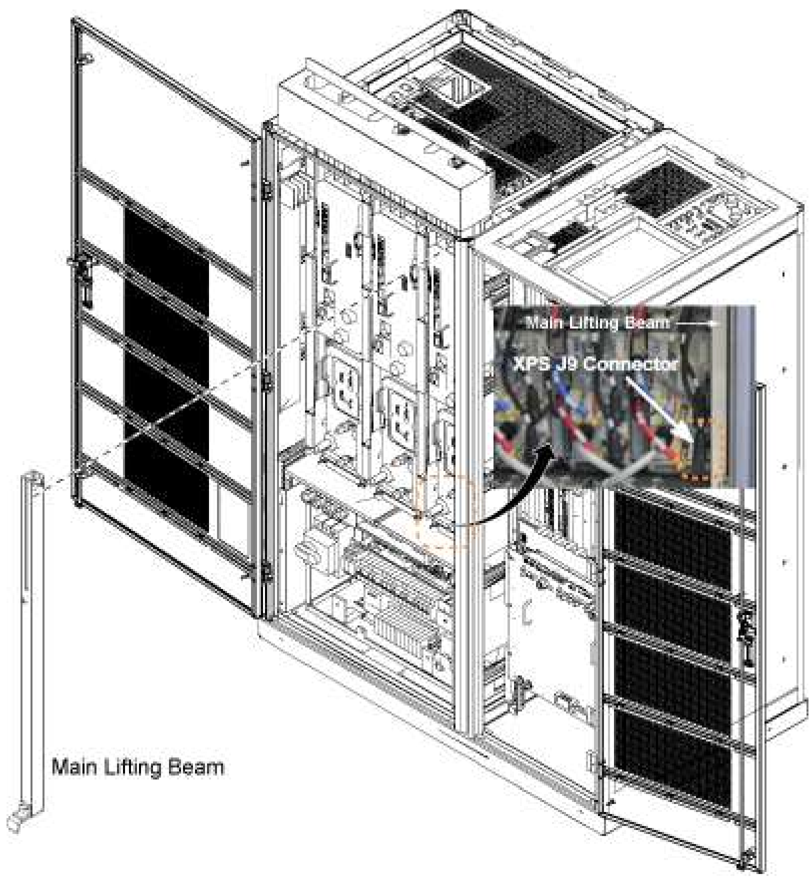
1. Power down the host PC.
2. The power in the cabinet needs to be OFF. Perform *LOTO for the PGR PDU/Gradient Subsystem* before the Dual Drive RF Amp is replaced.
3. Before disconnecting any coolant lines on the front of the Dual Drive RF Amp unit, shut down the Power electronics Pump (PE) at the Heat Exchange Cabinet (HEC) by pressing ▲ and **F3** simultaneously. Then switch the PE circuit breaker to **OFF**

Illustration 2: Power electronics Pump OFF



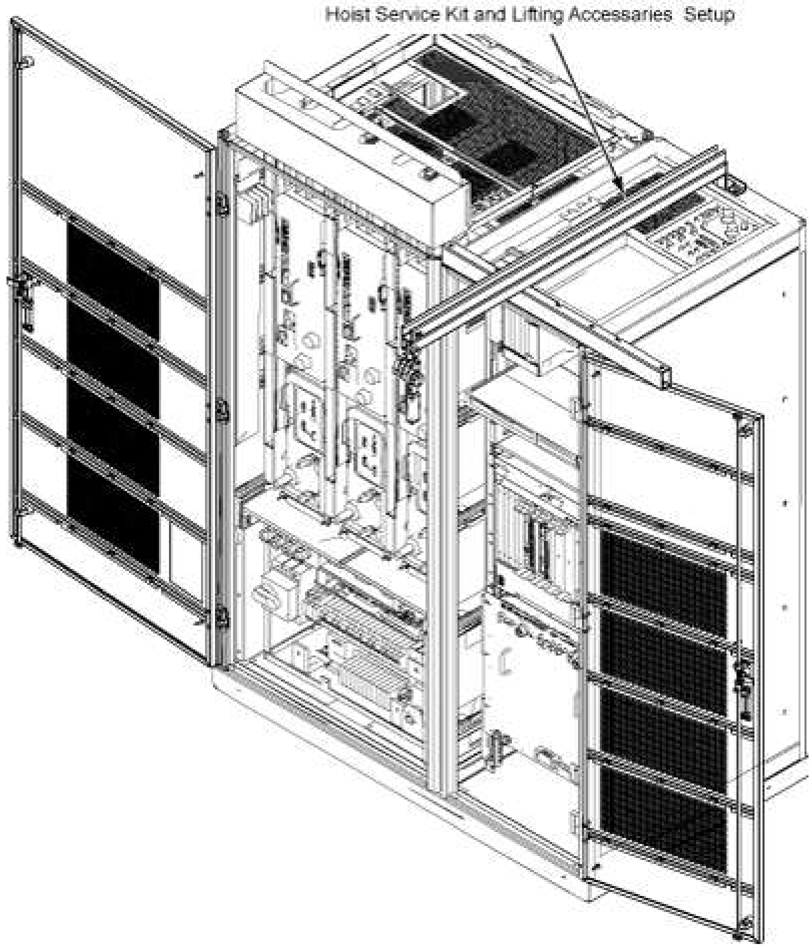
4. Before setting up the hoist kit, remove the eXtreme Gradient Power Supply (XPS) J9 connector to properly remove the main lifting beam. (Shown below.)

Illustration 3: Lifting Beam and Connector in PGR Cabinet



5. Perform the *Hoist Service Kit and Lifting Accessories* procedure to set up and secure the hoist.

Illustration 4: Hoist Kit Setup



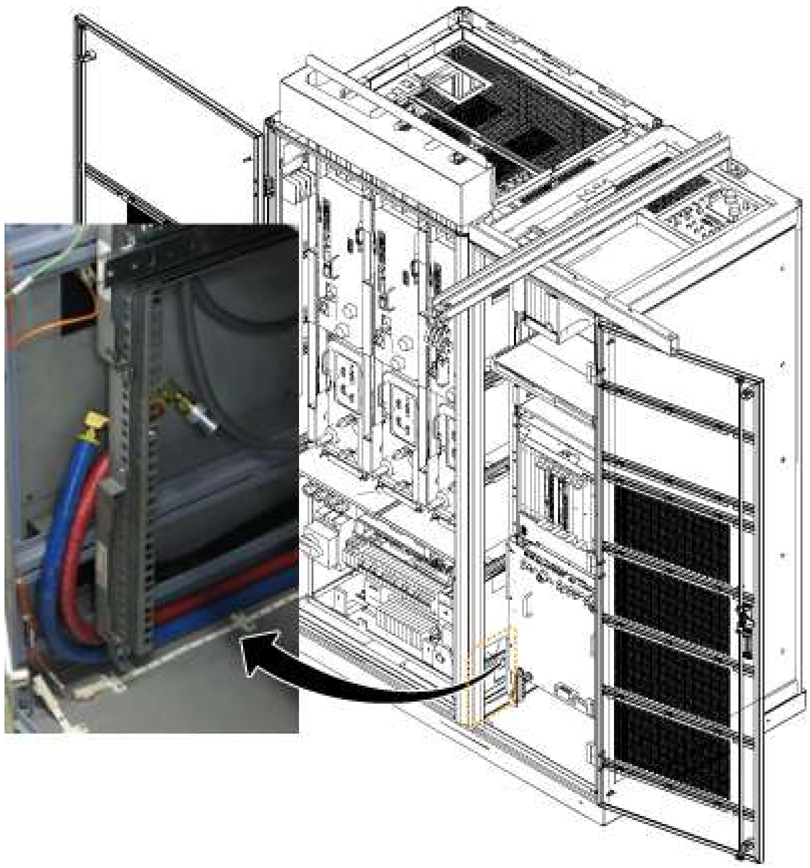
NOTE: Prior to the next step, prepare to absorb any leakage from the coolant lines with a towel in hand as the line is connected. Do not allow leakage to fall on white leak sensor strips located in cabinet in located in front and under RF amplifier.

6. Disconnect the cables on the front of the Dual Drive RF Amp, and disconnect the coolant lines.

Illustration 5: Disconnect the cables

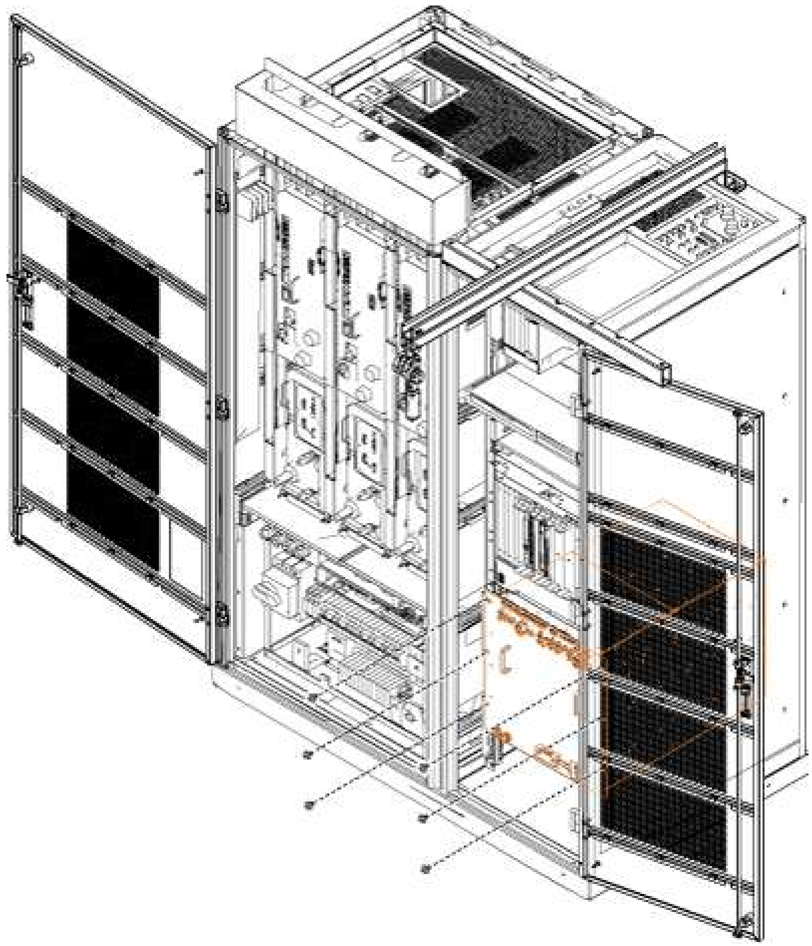
7. Carefully move the cables and lines to the side of the cabinet, allowing the surrounding area to be free from any obstruction of movement for when the amplifier will be pulled out. (Shown below.)

Illustration 6: Coolant Lines Placement



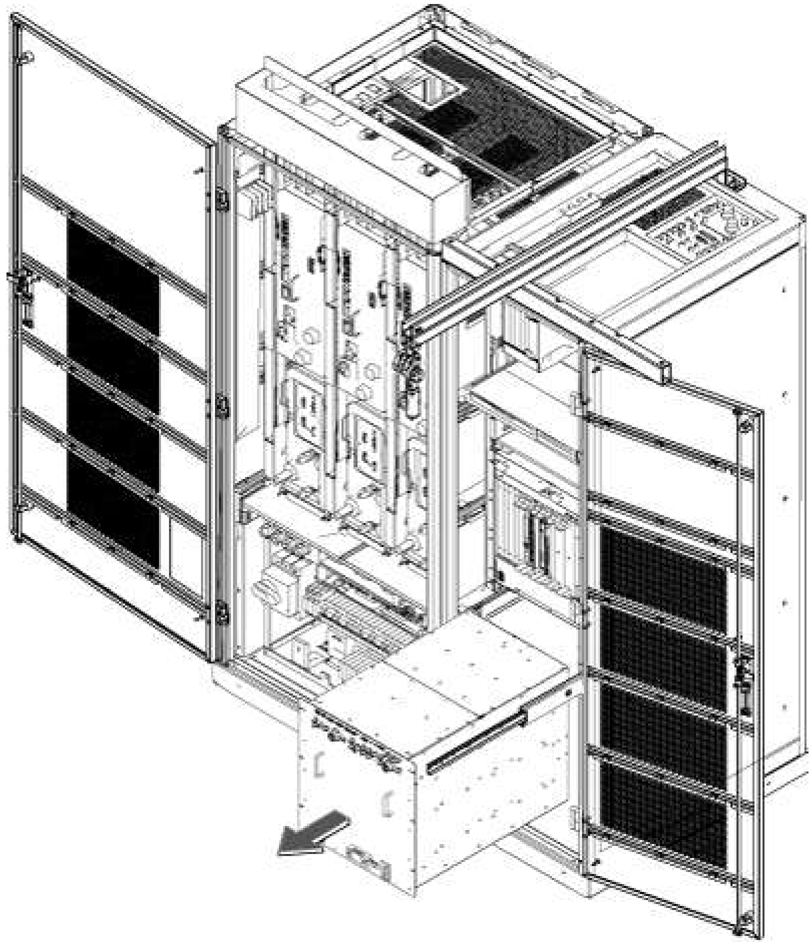
8. Unscrew the six attachment screws (shown below).

Illustration 7: Dual Drive RF Amp Attachment Screws



9. Using the two front handles, carefully slide out the Dual Drive RF Amp unit.

Illustration 8: Slide Out Dual Drive RF Amp



10. Attach the lifting brackets to both sides of the Dual Drive RF Amp. The lifting brackets can be found in the Dual Drive RF Amp FRU crate. (Shown below.) The attachment screws are captive screws and remain with the lifting brackets.

Illustration 9: Inside FRU Crate - Lifting Brackets

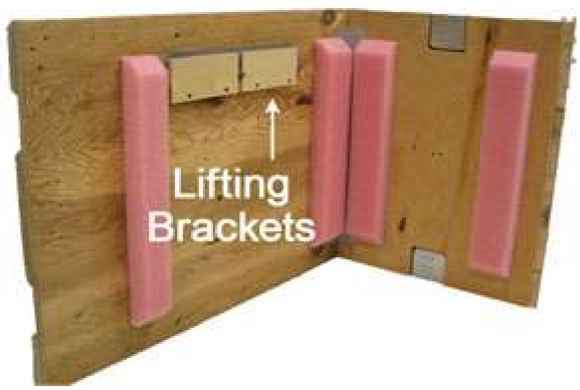
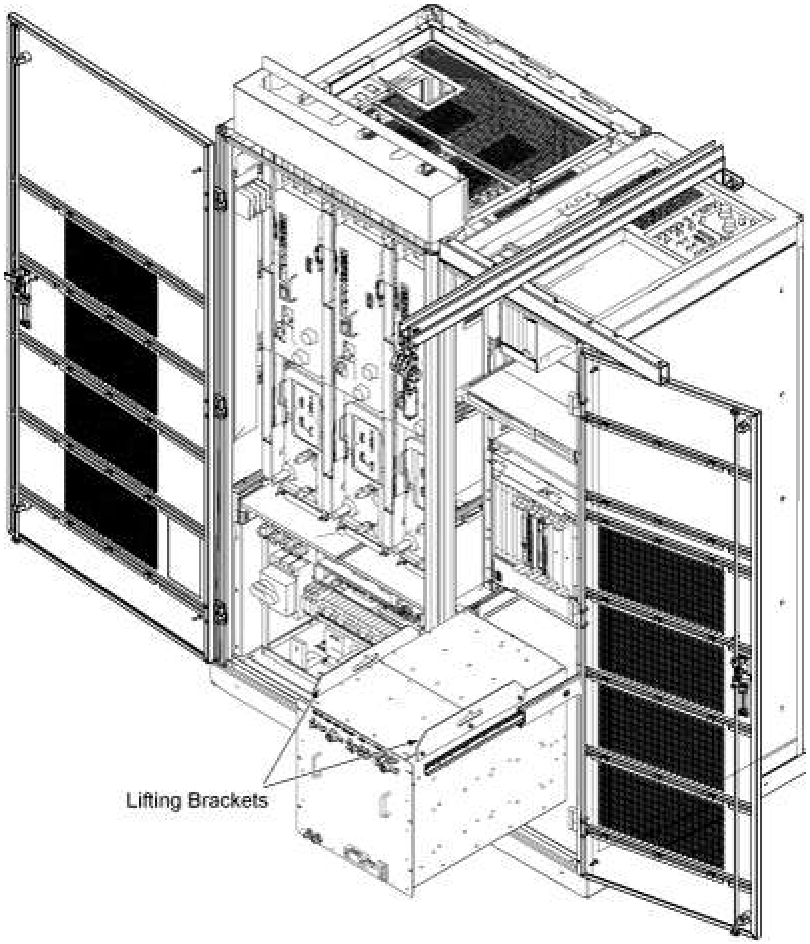


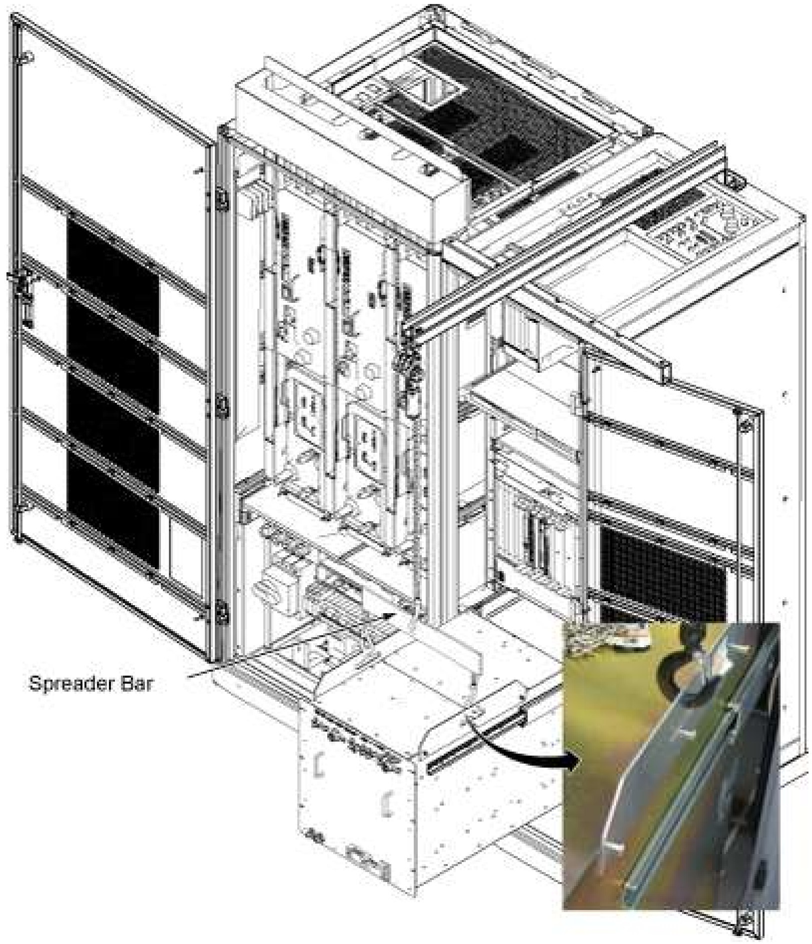
Illustration 10: Attach Lifting Brackets



11. Hook the winch and spreader bar from the hoist kit to the lifting bracket.

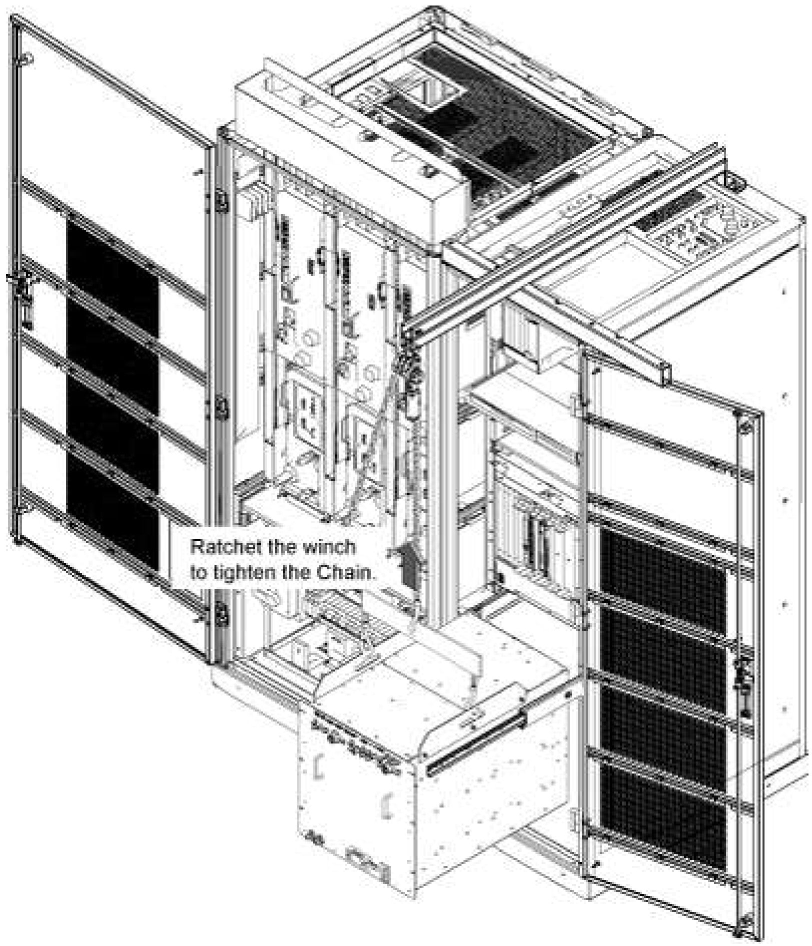
NOTE: Make sure the hook is seated in the lifting bracket. If not, the weight can shift causing harm to the equipment or engineer.

Illustration 11: Lifting Brackets with Seated Hook



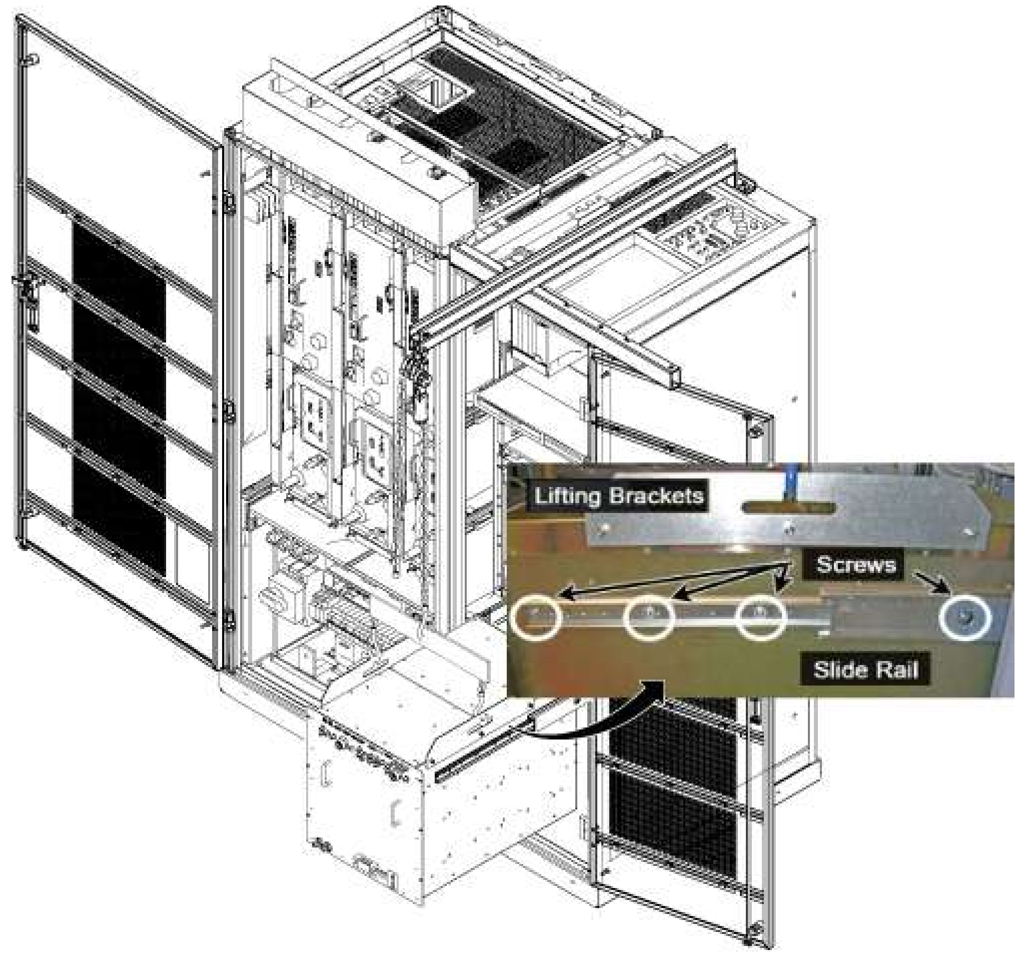
12. Ratchet the winch to tighten the chain.

Illustration 12: Tighten the chain



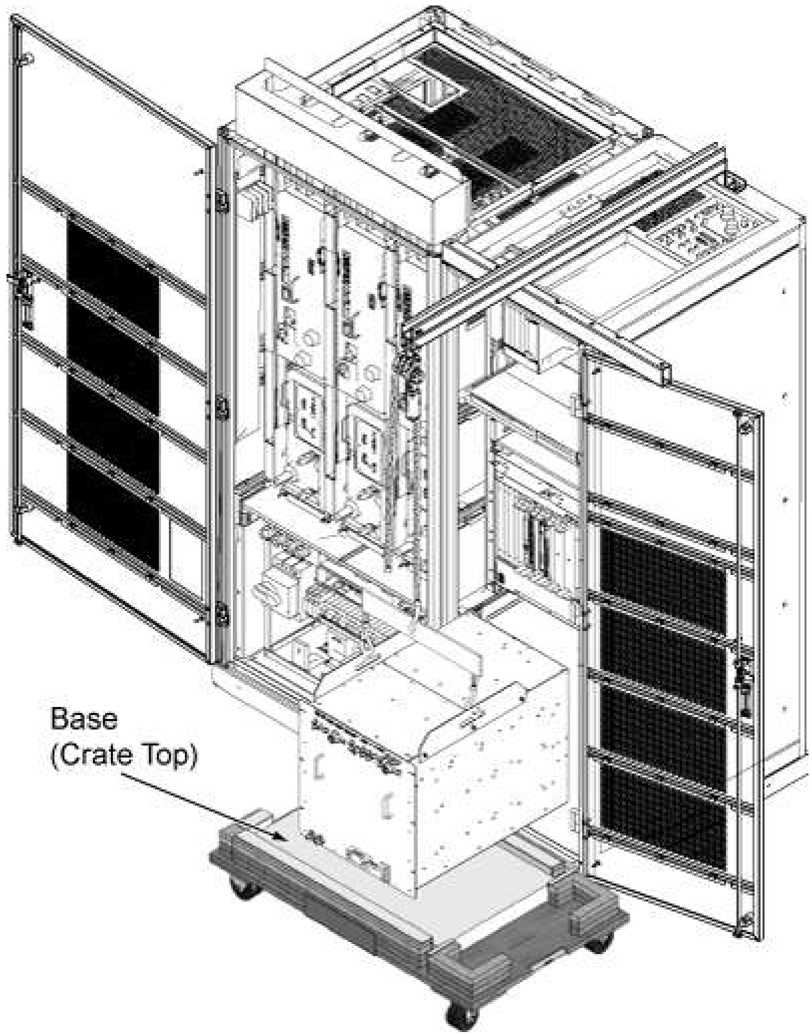
13. With the Dual Drive RF Amp unit supported by the hoist kit, unscrew the four screws and remove the slide rails from both sides. (Shown below.)

Illustration 13: Dual Drive RF Amp Slide Rails



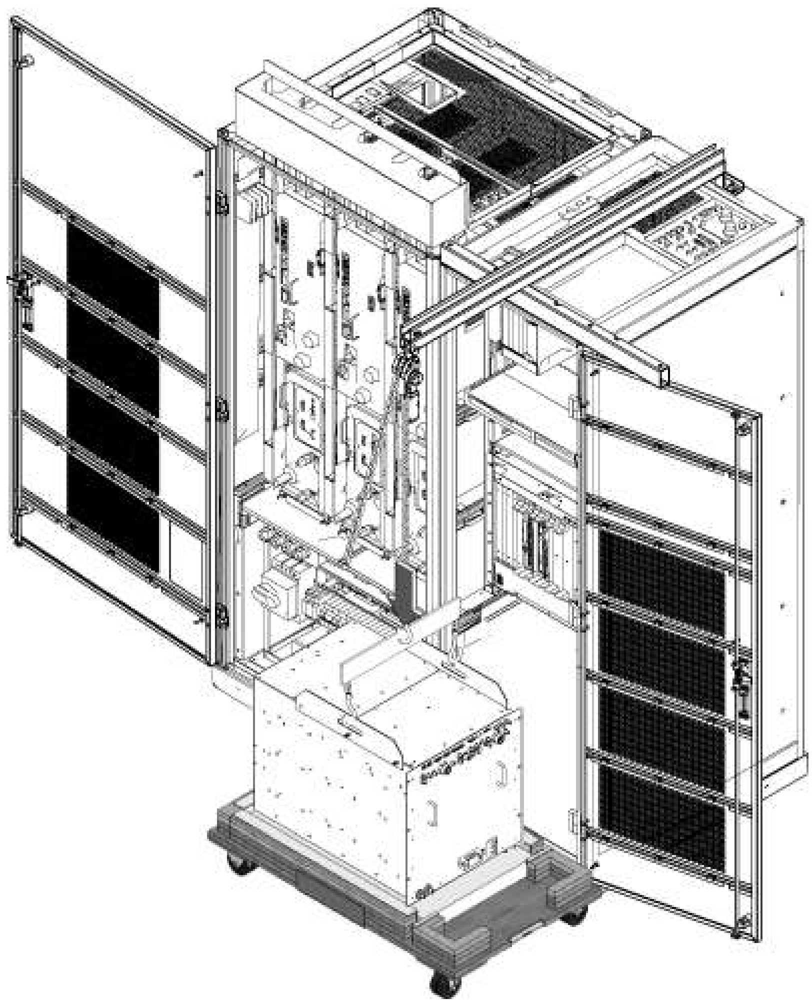
14. The FRU is packaged in a wooden crate. For the Dual Drive RF Amp, there are four parts to the FRU crate. Remove the top of the wooden crate and turn it on its base. The Dual Drive RF Amp being removed can be placed into the top. Roll the top underneath the Dual Drive RF Amp. (Shown below.) Lock the wheels so that the container does not move.

Illustration 14: Base of FRU Crate



15. Ratchet the winch to lower the Dual Drive RF Amp into the package.

Illustration 15: Lower the Dual Drive RF Amp



16. Remove the hook from the lifting bracket.

17. Unscrew the lifting brackets from the sides of the Dual Drive RF Amp.

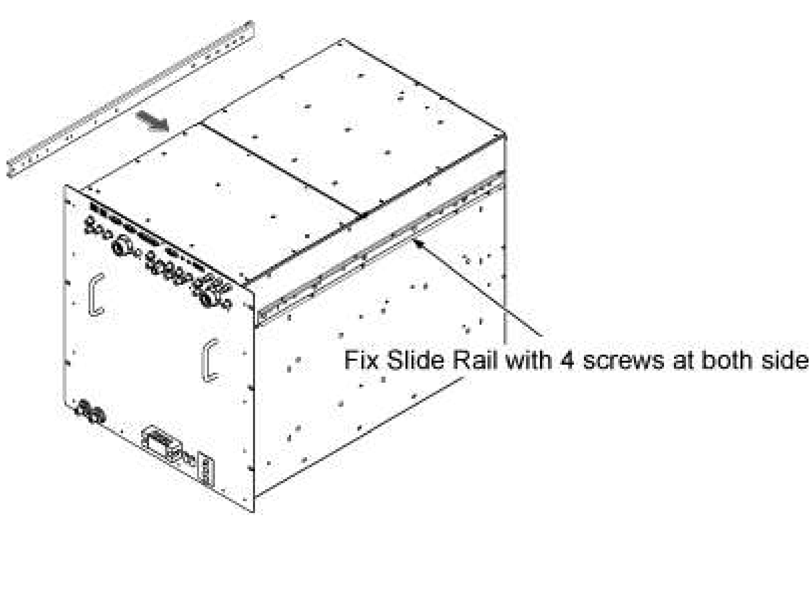
18. Roll the removed Dual Drive RF Amp to the side. Roll the removed Dual Drive RF Amp to the side as later it will need to be purged of water coolant prior repackaging for return.

19. Remove the center sections of the FRU crate.

20. Bring the FRU to the front of the cabinet.

21. Attach the slide rails to the FRU.

Illustration 16: Attach the slide rails

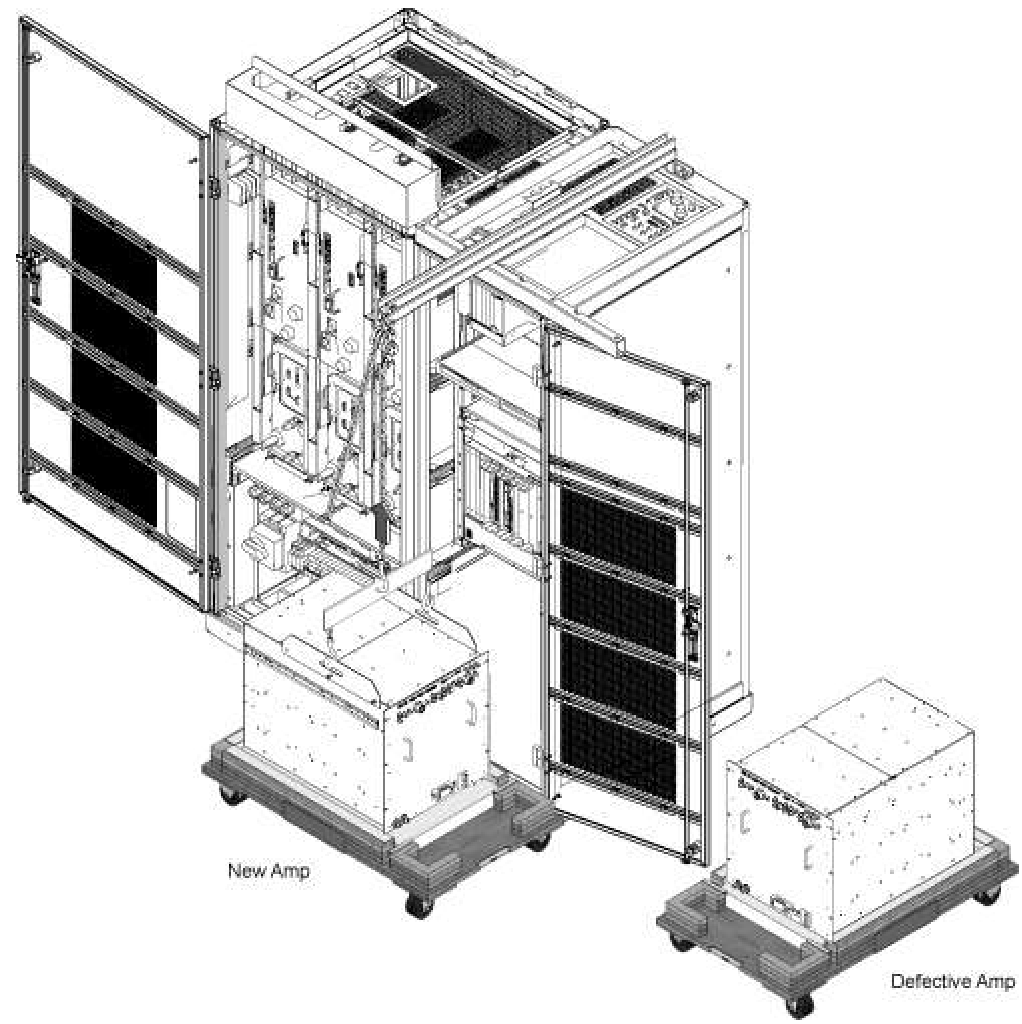


22. Attach the lifting brackets to the sides of the new amplifier.

23. Attach the new Dual Drive RF Amp lifting brackets to the winch and spreader bars.

NOTE: Make sure the hook is seated in the lifting bracket. If not the weight can shift causing harm to the equipment or the engineer.

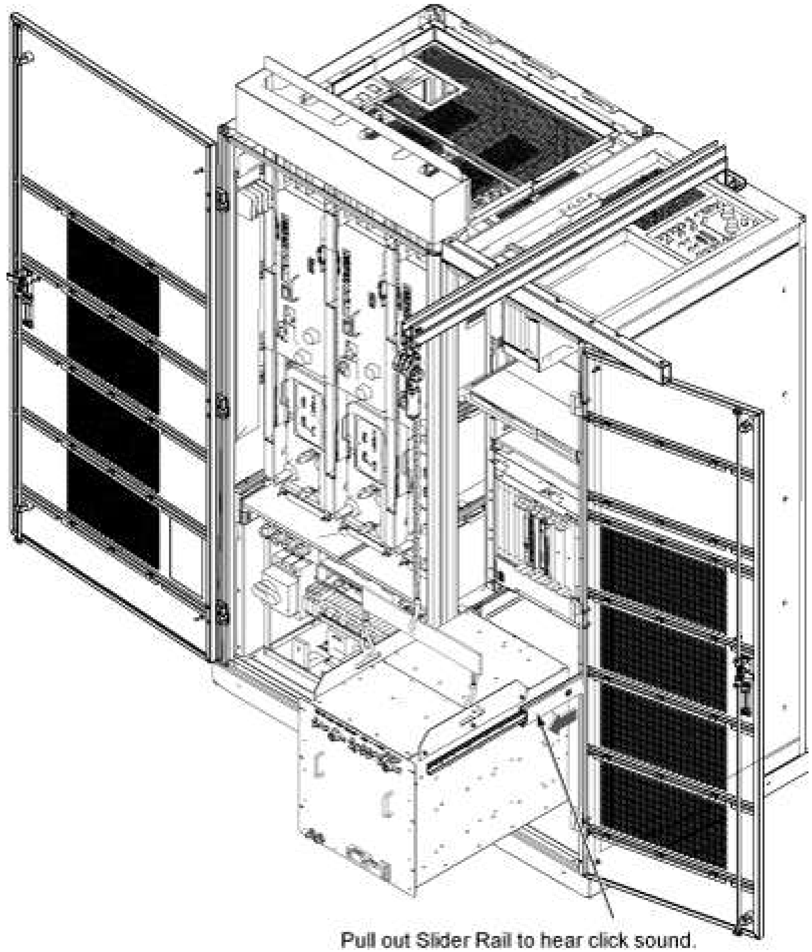
Illustration 17: Dual Drive RF Amp FRU



24. Ratchet the winch to tighten the chains and begin lifting the unit. Slide out the rail a short amount to gauge how high to lift the Dual Drive RF Amp. The Dual Drive RF Amp slide rails and the cabinet slide rails must align for correct insertion.

25. When the slide rails are aligned, pull the slide rails out a short distance to get each side started. Once both of the slide rails are attached, pull the slide rails onto the FRU until you hear a click. This click is a signal that the unit is docked in the rails. Ensure the Dual Drive RF Amp is secure on the rails.

Illustration 18: Restore Amp



26. Unhook the winch and spreader bars from the lifting brackets. Remove the lifting brackets from the sides of the new Dual Drive RF Amp.

    Place the lifting brackets into the Dual Drive RF Amp FRU crate.

27. Slide the Dual Drive RF Amp into the cabinet completely.

28. Attach the Dual Drive RF Amp to the cabinet using the front six attachment screws (see Illustration 7).

NOTE: Prior to the next step, prepare to absorb any leakage from the coolant lines with a towel in hand as the line is connected. Any coolant leakage on white leak sensor strips located in cabinet under RF amplifier must be thoroughly dried in order to prevent trips of PDU rotary main breaker by leak detection circuitry after application of PDU power.

29. Reconnect the remaining cables and coolant lines to the Dual Drive RF Amp.

30. Disassemble the *hoist kit*. Return the main lifting beam to the PGR cabinet. Once the beam is secure, reconnect the XPS J9 connector. Return the support tube to the HEC.

31. Proceed to Finalization, Step 1.

5 Finalization

1. Restore the power using the *LOTO for the PGR PDU/Gradient Subsystem*.

2. Power up the host PC.

3. Turn on the PPMP CB at the HEC. Start the pump by pressing ▲ and **F3** simultaneously.

4. *Calibrate the Dual Drive RF Amp*.

5. Perform *Universal Power Monitor - Head and Body Setup and Calibration* .

6. Perform UPM - Body and Head Functional Check .
7. Perform DD Quadrature calibration.
8. In order to prevent freeze damage to internal coolant pipes during return shipment, remove water coolant from the Dual Drive RF Amp unit using the manual coolant removal tool (see Coolant Draining).
9. After attaching the center sections of the FRU package to the removed FRU unit base, attach the top of the FRU crate to prepare the FRU for return.

   NOTE: The Dual Drive RF Amp has two center sections.

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